

Digital Image Processing

Homework 2

PART1:

2.12* Suppose that a flat area with center at (x_0, y_0) is illuminated by a light source with intensity distribution

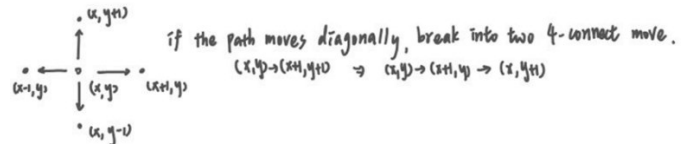
$$i(x, y) = Ke^{-[(x-x_0)^2 + (y-y_0)^2]}$$

Assume for simplicity that the reflectance of the area is constant and equal to 1.0, and let $K = 255$. If the intensity of the resulting image is quantized using k bits, and the eye can detect an abrupt change of $\frac{1}{8}$ intensity levels between adjacent pixels, what is the highest value of k that will cause visible false contouring?

$$\begin{aligned} f(x, y) &= i(x, y) \cdot R(x, y) \\ &= 255 e^{-[(x-x_0)^2 + (y-y_0)^2]} (1, 0) \\ &= 255 \cdot e^{-[(x-x_0)^2 + (y-y_0)^2]} \end{aligned}$$

$$8 \leq \frac{255+1}{2^k}, \quad 2^k \leq 32, \quad k \leq 5$$

2.16 Develop an algorithm for converting a one-pixel-thick m -path to a 4-path.

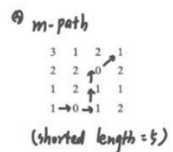
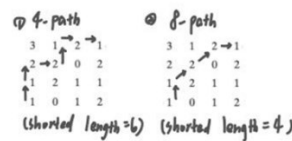
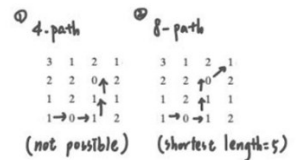


2.18 Consider the image segment shown in the figure that follows.

(a)* As in Section 2.5, let $V = \{0, 1\}$ be the set of intensity values used to define adjacency. Compute the lengths of the shortest 4-, 8-, and m -path between p and q in the following image. If a particular path does not exist between these two points, explain why.

	3	1	2	1	q
	2	2	0	2	
	1	2	1	1	
p	1	0	1	2	

(b) Repeat (a) but using $V = \{1, 2\}$.



2.37 We know from Eq. (2-45) that an affine transformation of coordinates is given by

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \mathbf{A} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

where (x', y') are the transformed coordinates, (x, y) are the original coordinates, and the elements of $\mathbf{A}$ are given in Table 2.3 for various types of transformations. The inverse transformation, $\mathbf{A}^{-1}$, to go from the transformed back to the original coordinates is just as important for performing inverse mappings.

- Find the inverse scaling transformation.
- Find the inverse translation transformation.
- Find the inverse vertical and horizontal shearing transformations.
- Find the inverse rotation transformation.
- Show a composite inverse translation/rotation transformation.

(a) scaling⁻¹:

$$\begin{bmatrix} C_x & 0 & 0 \\ 0 & C_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad x' = C_x v, \quad y' = C_y w$$

(b) translation⁻¹:

$$\begin{bmatrix} 1 & 0 & T_x \\ 0 & 1 & T_y \\ 0 & 0 & 1 \end{bmatrix} \quad x' = v + T_x, \quad y' = w + T_y$$

(c) shearing⁻¹:

$$\begin{bmatrix} 1 & S_x & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad x' = v + S_x w, \quad y' = w$$

$$\begin{bmatrix} 1 & 0 & 0 \\ S_y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad x' = v, \quad y' = S_y x + w$$

(d) rotation⁻¹:

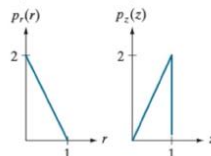
$$\begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad x' = v \cos \theta - w \sin \theta$$

$$\quad y' = v \sin \theta + w \cos \theta$$

(e) translation/rotation:

$$\mathbf{R} * \mathbf{T} = \begin{bmatrix} \cos \theta & -\sin \theta & T_x \cos \theta - T_y \sin \theta \\ \sin \theta & \cos \theta & T_x \sin \theta + T_y \cos \theta \\ 0 & 0 & 1 \end{bmatrix}$$

3.12 An image with intensities in the range $[0, 1]$ has the PDF, $p_r(r)$, shown in the following figure. It is desired to transform the intensity levels of this image so that they will have the specified $p_z(z)$ shown in the figure. Assume continuous quantities, and find the transformation (expressed in terms of r and z) that will accomplish this.



$$\int_0^r p_r(w) dw = \int_0^r (2 - 2w) dw = -r^2 + 2r$$

$$\int_0^z p_z(w) dw = \int_0^z 2w dw = z^2$$

$$z^2 = -r^2 + 2r$$

$$\Rightarrow z = \sqrt{-r^2 + 2r}$$

(positive level)

3.18 You are given the following kernel and image:

$$w = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix} \quad f = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

(a) $w = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$

overlapping $f = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

(a)* Give a sketch of the area encircled by the large ellipse in Fig. 3.28 when the kernel is centered at point (2,3) (2nd row, 3rd col) of the image shown above. Show specific values of w and f .

(b)* Compute the convolution $w * f$ using the minimum zero padding needed. Show the details of your computations when the kernel is centered on point (2,3) of f and then show the final full convolution result.

(c) Repeat (b), but for correlation, $w \circ f$.

(c)

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix} = 6$$

no need to flip the kernel

$\Rightarrow$ same result as (b)

(b)

$$g(x, y) = w(-1, -1) f(x-1, y-1) + w(-1, 0) f(x-1, y) + w(0, -1) f(x, y-1) + w(0, 0) f(x, y) + w(1, -1) f(x+1, y-1) + w(1, 0) f(x+1, y) + w(2, 0) f(x+2, y)$$

$$= \sum_{s=-A}^A \sum_{t=-B}^B w(s, t) f(x+s, y+t)$$

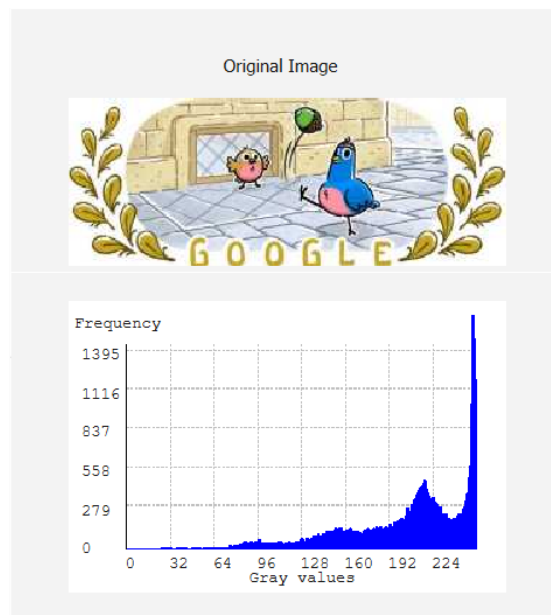
$$f' = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

$$= 0 + 0 + 0 + 0 + 4 + 0 + 0 + 2 + 0 = 6$$

flipped kernel

PART2: Image File Reading, Display and Basic Processing

1. Read a color image file and display it on the screen.



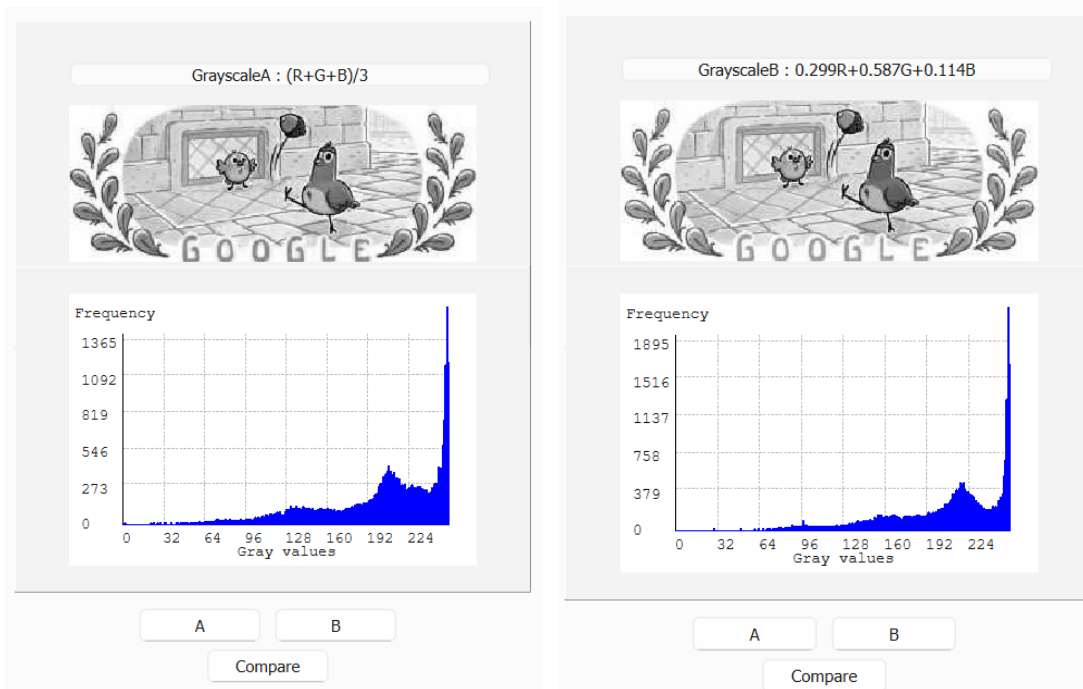
2. Convert a color image into a grayscale image using the following equations:

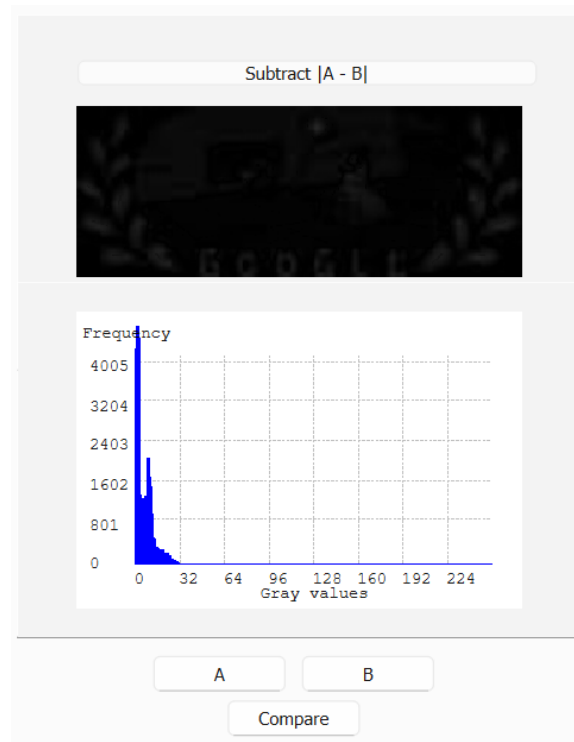
A. $GRAY = (R+G+B)/3.0$

B. $GRAY = 0.299*R + 0.587*G + 0.114*B$

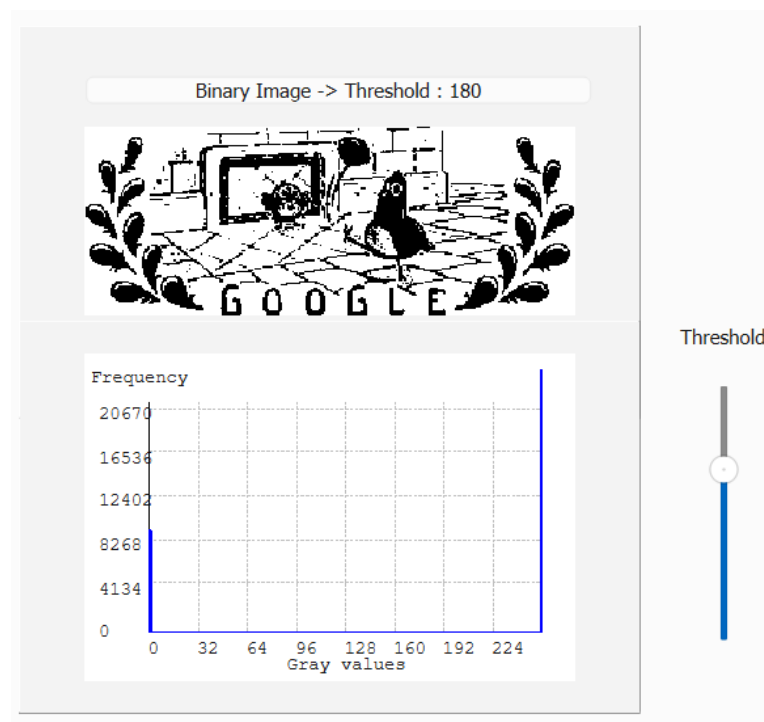
Compare the difference between two images by image subtraction.

Determine and display the histogram of a grayscale image.

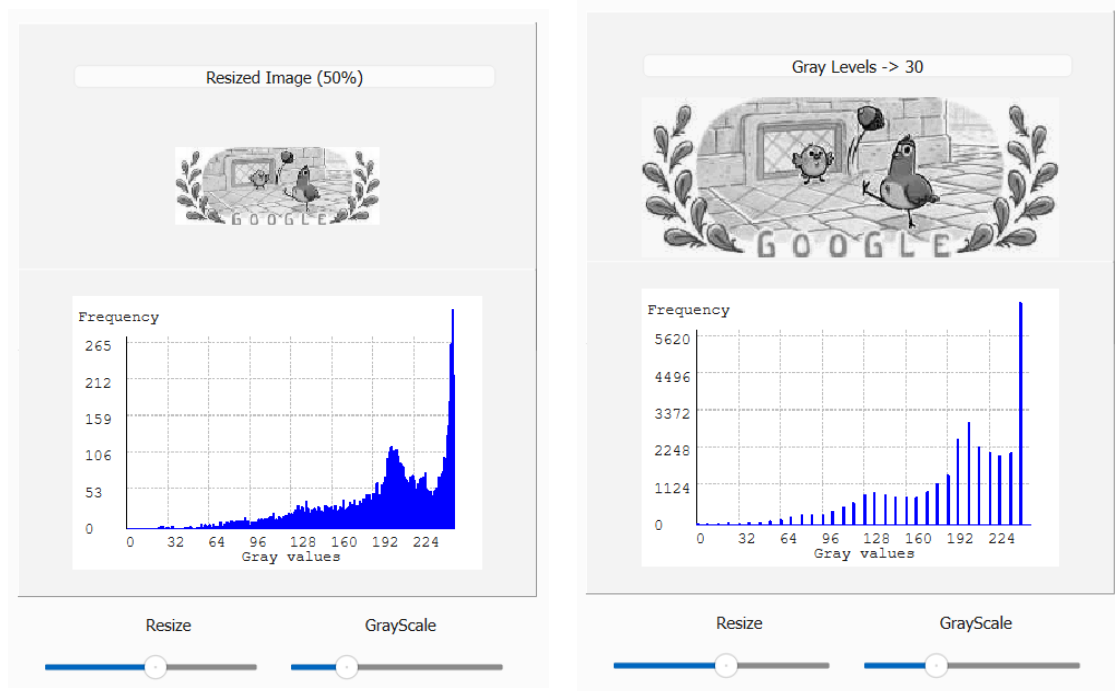




- **grayA**: Simple average method where the red, green, and blue values are averaged.
 - **grayB**: Weighted average based on human perception (0.299 for red, 0.587 for green, 0.114 for blue).
3. Implement a manual threshold function to convert a grayscale image into a binary image.

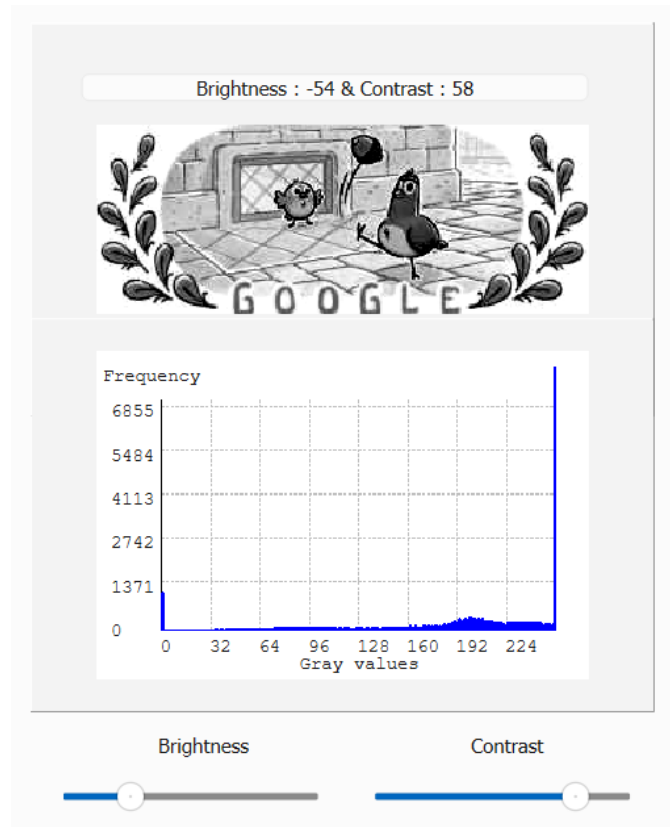


4. Implement a function to adjust the spatial resolution (enlarge or shrink) and grayscale levels of an image. Use an interpolation method on enlarging an image.



function is used for **resampling** the image, which can be useful for zooming in and out or adapting the image size for different display requirements.

5. Implement a function to adjust the brightness and contrast of an image.

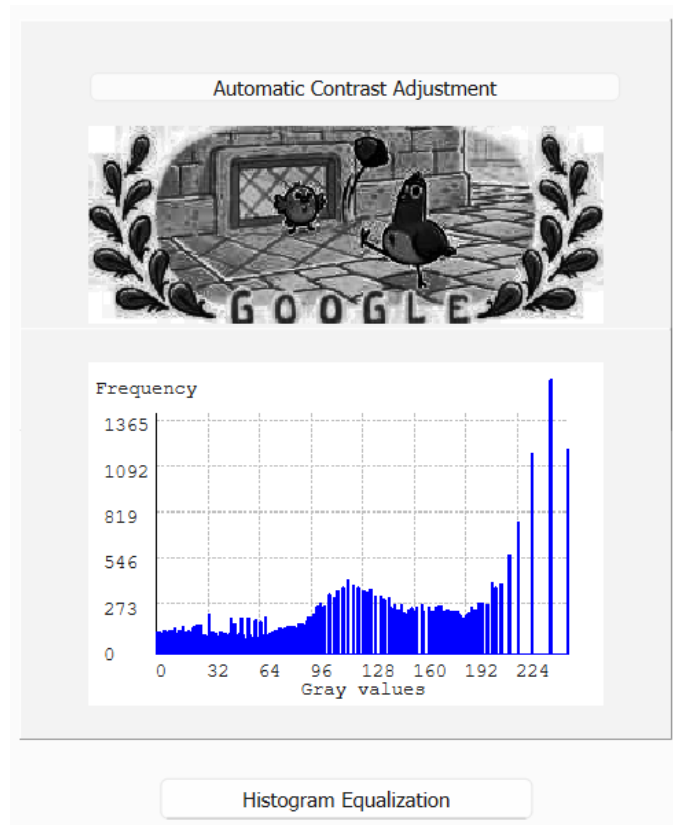


For each pixel $I(x,y)$, the transformation can be expressed as:

$$I' = \text{contrast} \times (I - 128) + 128 + \text{brightness}$$

Where I' is the new pixel value, I is the original pixel value, and 128 is the midpoint of the grayscale range (for 8-bit images).

- Implement a histogram equalization function for automatic contrast adjustment.



- **Histogram Calculation:** The function first computes the frequency of each intensity value (0-255) in the grayscale image.
- **Cumulative Distribution Function (CDF):** The CDF is calculated based on the histogram, which helps in redistributing pixel values across the entire intensity range.
- **Pixel Value Adjustment:** Each pixel's intensity is adjusted according to the CDF, spreading out the pixel values more evenly across the grayscale spectrum.

Compare to the default photo editor in ios system :

